

Maximizing Fisher Information in ϵ -Differential Privacy Mechanisms

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Massive amount of personal data collected daily, growing in size in the modern era of big data. Some examples include:

- smart phone and smart wearable devices data
- smart meter data
- medical research
- government statistics

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Figure 1 is a line graph showing the power consumption of a house over a 24-hour period. The y-axis represents power in Watts, ranging from 0 to 9000. The x-axis represents time in hours, from 00 to 23. The graph shows several peaks in power consumption. A large peak occurs around 09:00, labeled 'Breakfast' and 'Water Heater'. Another peak occurs around 13:00, labeled 'Water Heater'. A third peak occurs around 19:00, labeled 'Evening Activities: Dinner, Showers, Laundry, Working on Computer' and 'Shower'. A fourth peak occurs around 23:00, labeled 'Shower'. The graph also shows a period of low power consumption from 00:00 to 07:00, labeled 'Overnight Period'. A small peak around 04:00 is labeled 'Refrigerator'.

Navigation icons: back, forward, search, and other controls.

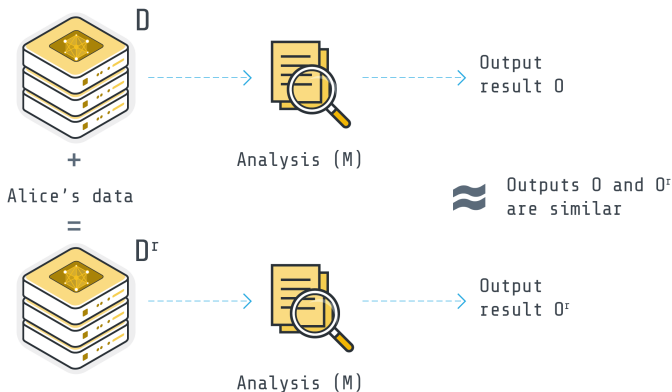
Traditional solutions tried to anonymize or aggregate the data before public releases. "Reconstruction attacks" exposed drawbacks to such simple solutions

- 2010 U.S. Census data reconstruction attack - successful recovery of individual microdata¹
- Netflix Prize dataset - researchers used publicly available IMDb reviews to re-identify individuals in the dataset²

¹T. Dick, C. Dwork, M. Kearns, T. Liu, A. Roth, G. Vietri, & Z.S. Wu, Confidence-ranked reconstruction of census microdata from published statistics, Proc. Natl. Acad. Sci. U.S.A. 120 (8) e2218605120, <https://doi.org/10.1073/pnas.2218605120> (2023).

²A. Narayanan and V. Shmatikov, "Robust De-anonymization of Large Sparse Datasets," 2008 IEEE Symposium on Security and Privacy (sp 2008), Oakland, CA, USA, 2008, pp. 111-125, doi: 10.1109/SP.2008.33. 

Individual records shouldn't affect the overall statistics too much.



- For a given private data set $X = (X_1, \dots, X_n)$ in some input space $\mathcal{X}^n$ randomly generate sanitized (protected) data $Z \in \mathcal{Z}$ with conditional distribution

$$Q(A|x) = P(Z \in A | X = x).$$

- The conditional distribution (Markov kernel) Q is called a *privacy mechanism*
- The distribution of the sanitized data Z is given by

$$[QP_\theta](A) := \int_{\mathcal{X}} Q(A|x) P_\theta(dx).$$

- Imposes a requirement on the ratio of randomization probabilities:

$$e^{-\epsilon} \leq \frac{Q(z|x)}{Q(z|x')} \leq e^{\epsilon} \quad \forall x, x' \in \mathcal{X}, \forall z \in \mathcal{Z}.$$

- The privacy level is characterized by a parameter $\epsilon \in [0, \infty)$
- Ensures that no single record can influence the database (and subsequently the resulting statistics) too much.

- Common Setup: $X_1, \dots, X_n \sim P_\theta$

Global Differential Privacy

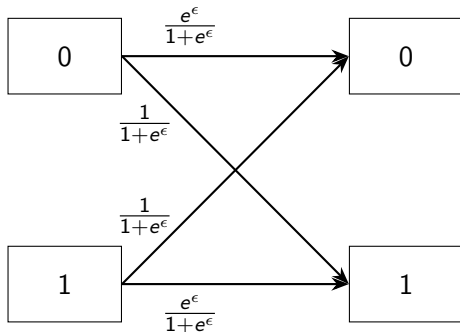
- Central data curator that collects the data and releases statistics
- Statistics first calculated, noise is added to the result
- $\hat{\theta}_n = f(X_1, \dots, X_n) + \text{Noise}$

Local Differential Privacy

- No central data curator, randomization happens on the client level
- Statistics are estimates of noisy protected data
- $Z_i = X_i + \text{Noise}, \quad \hat{\theta}_n = f(Z_1, \dots, Z_n)$

Input Alphabet $\mathcal{X}$

Output Alphabet $\mathcal{Z}$



$$\frac{\Pr[Z=j|X=0]}{\Pr[Z=j|X=1]} \leq e^\epsilon$$

For all $j \in \{0, 1\}$

- However, an increase in privacy reduces the utility of the data
- Utility is measured through a utility function
- Fisher Information, f-divergence, entropy, ...
- Consider extreme examples of $\epsilon = 0$ (maximum privacy) and $\epsilon = \infty$ (no privacy)

- Let X be a random variable with PDF $p_\theta(x)$ depending on a parameter θ
- Fisher information measures how much information X carries about θ :

$$I_\theta = E_\theta \left[\left(\frac{\partial}{\partial \theta} \log p_\theta(X) \right)^2 \right]$$

- An unbiased estimator $\hat{\theta}$ is efficient if it achieves the Cramér-Rao bound for every θ

$$\text{Var}_\theta(\hat{\theta}) \geq \frac{1}{I_\theta}$$

- Unbiased estimator: $E_\theta[\hat{\theta}] = \theta, \quad \forall \theta \in \Theta$

- Private data (prior distribution) is iid generated by some statistical model parameterized by θ : $X_1, \dots, X_n \sim P_\theta$
- Sanitized data, $Z_1, \dots, Z_n$ generated through the privatization mechanism Q

$$Z = QP_\theta$$

- Given ϵ -differential privacy, there exist many mechanisms that generate private data and satisfy ϵ -differential privacy constraints:

$$D_\epsilon(\mathcal{X}) = \bigcup_{(Z, \mathcal{G})} \{Q \in M(\mathcal{X} \rightarrow \mathcal{Z}) \mid Q(A|x) \leq e^\epsilon Q(A|x'), \quad \forall A, x, x'\}$$

- We want to maximize the Fisher information of the protected data

$$\sup_{Q \in D_\epsilon(\mathcal{X})} I_\theta(Q)$$

Continuous prior distributions

- infinite-dimensional domain $D_\epsilon(\mathcal{X})$

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$$\sup_{Q \in D_\epsilon(\mathcal{X})} I_\theta(Q) = \sup_{Q \in D_\epsilon(\mathcal{X})} E_\theta \left[\left(\frac{\partial}{\partial \theta} \log[QP_\theta](Z) \right)^2 \right]$$

- Perhaps possible for certain classes of distributions, calculus of variations

Discrete prior distributions, $|\mathcal{X}| = k \in \mathbb{N}$

- suffices to limit the optimization domain to

$$\begin{aligned} D_{\epsilon,k} &= \bigcup_{\mathcal{Z}: |\mathcal{Z}|=k} \{Q \in \mathcal{M}(\mathcal{X} \rightarrow \mathcal{Z}) \mid Q(A|x) \leq e^\epsilon Q(A|x') \quad \forall A, x, x'\} \\ &= \{Q \in [0, 1]^{k \times k} \mid \sum_{i=1}^k Q_{ij} = 1, Q_{ij} \leq e^\epsilon Q_{ij'} \quad \forall i, j, j'\} \end{aligned}$$

- (relatively) solvable maximization problem

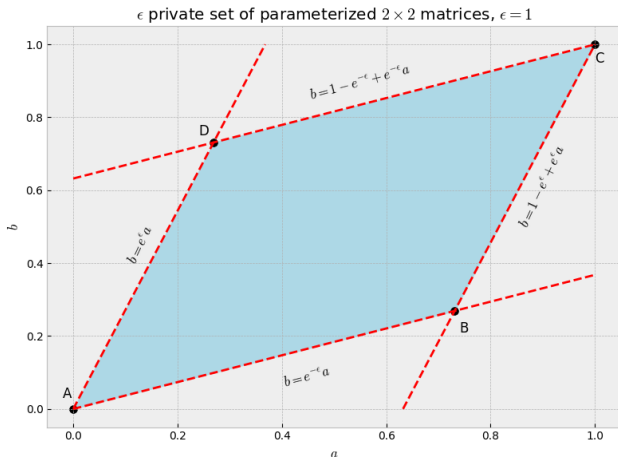
$$\max_{Q \in D_{\epsilon,k}} I_\theta(Q) = \max_{Q \in D_{\epsilon,k}} \sum_{z \in \mathcal{Z}} \frac{(\sum_{x \in \mathcal{X}} Q(z|x) \dot{p}_\theta(x))^2}{\sum_{x \in \mathcal{X}} Q(z|x) p_\theta(x)}$$

- Kairouz et al. (2016) give an equivalent linear programme with exponential time complexity $O(2^k)$

Can we do better than exponential time complexity?

- Optimization domain $D_{\epsilon,k}$ is convex
- Objective function $I_{\theta}(Q)$ is convex in Q and differentiable with respect to Q
- Convex maximization problem over a convex set
- Potential local maxima

Consider $|\mathcal{X}| = |\mathcal{Z}| = k = 2$, $Q = \begin{pmatrix} a & b \\ 1-a & 1-b \end{pmatrix}$, $0 \leq a, b \leq 1$



- Objective function

$$l_{\theta}(Q) = \sum_{z \in \mathcal{Z}} \frac{(Q_z \cdot \dot{p}_{\theta})^2}{Q_z \cdot p_{\theta}}$$

- Gradient

$$\nabla l_{\theta}(Q) \big|_{(x,z)} = 2\dot{p}_{\theta}(x) \frac{Q_z \cdot \dot{p}_{\theta}}{Q_z \cdot p_{\theta}} - p_{\theta}(x) \frac{(Q_z \cdot \dot{p}_{\theta})^2}{(Q_z \cdot p_{\theta})^2}$$

- Constraint set is a high dimensional convex polytope, with $k \cdot (k - 1)$ ϵ -privacy constraints per row, plus k column stochasticity constraints for a total of $k^2 \cdot (k - 1) + k$ constraints

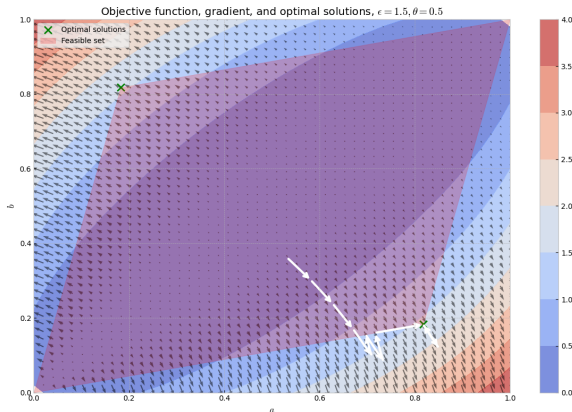
$$Q_{ij} \leq e^{\epsilon} Q_{ij'}, \sum_{i=1}^k Q_{ij} = 1 \quad \forall i, j, j'$$

- Update step

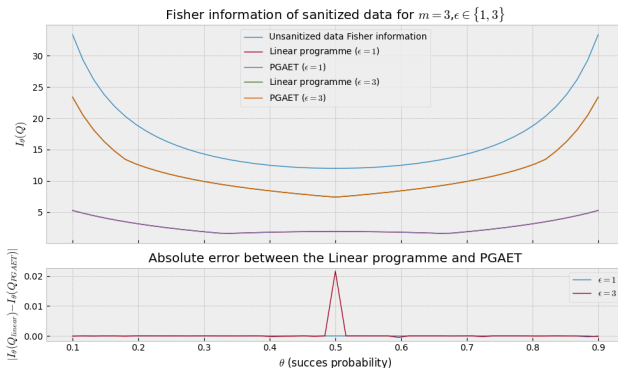
$$Q_{t+1} = P_{D_{\epsilon,k}}(Q_t + \lambda_t \nabla l_{\theta}(Q_t))$$

- $P_{D_{\epsilon,k}}(Q) = \arg \min \{ \|Q - Z\|_F : Z \in D_{\epsilon,k} \}$ projection subroutine
- Projection subroutine can be formulated as a quadratic programme, with time complexity of $O(k^9)$
- Overall complexity of the algorithm is limited by the projection subroutine
- Periodic mapping of the boundary and subsequent larger steps along the boundary
- Problem of local maxima alleviated with multiple restarts
- We term the algorithm Projected Gradient Ascent with Edge Traversal (PGAET)

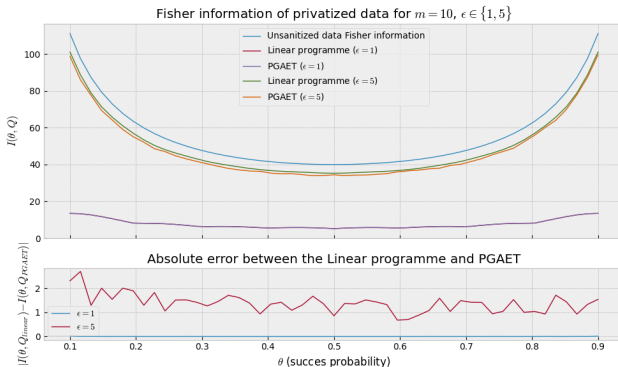
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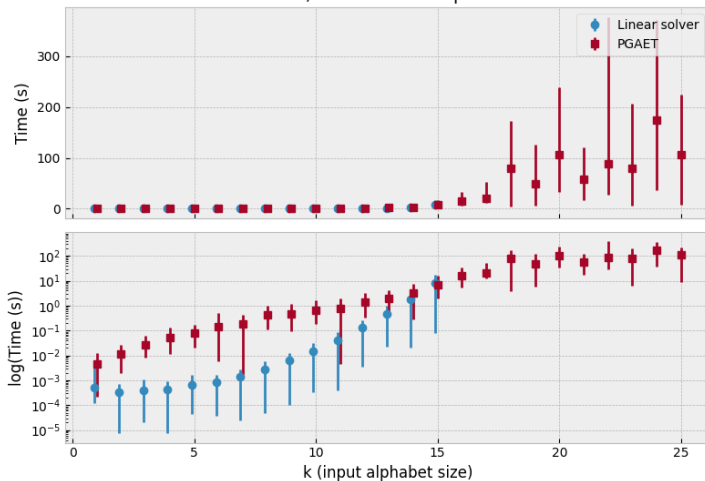
Let $p_{\theta}^{(m)}(x) = \binom{m}{x} \theta^x (1 - \theta)^{m-x}$, $x \in \{0, \dots, m\}$



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Runtime comparisons, $\theta = 0.5, \epsilon = 1.0$
median, 10th and 90th percentile



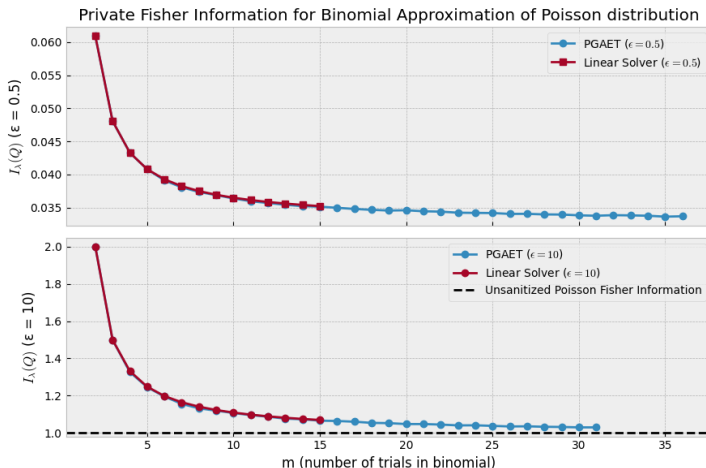
- Probability density function: $p_\lambda(x) = \frac{e^{-\lambda} \lambda^x}{x!}$, $x \in \mathbb{N}_0$
- Fisher information: $I_P(\lambda) = \frac{1}{\lambda}$
- Approximate the Poisson distribution with the Binomial

$$p_\lambda(x) \approx p_{\theta=\frac{\lambda}{m}}^{(m)}(x) = \binom{m}{x} (\lambda/m)^x (1 - \lambda/m)^{1-x}, \quad x \in \{0, \dots, m\}$$

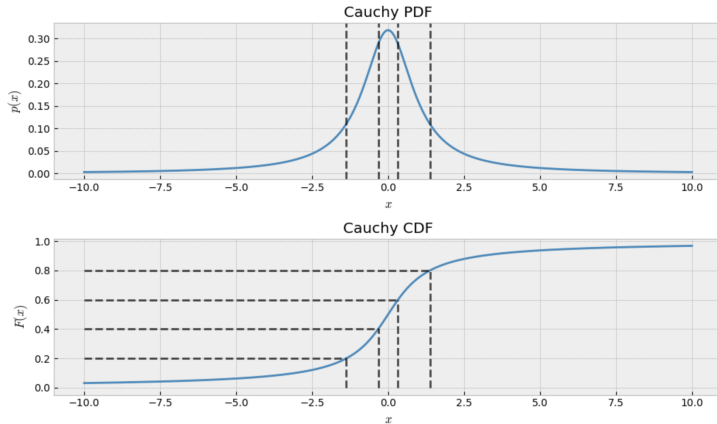
- Fisher information of λ in the binomial formulation:

$$I_B^{(m)}(\lambda) = \frac{1}{\lambda} + \frac{1}{m} + \frac{\lambda}{m(m - \lambda)}$$

- $\lim_{m \rightarrow \infty} I_B^{(m)}(\lambda) = I_P(\lambda)$



- Consider data from the Cauchy distribution $X_1, \dots, X_n \sim \text{Cauchy}(\theta)$
- Probability density function $p_\theta(x) = \frac{1}{\pi[1+(x-\theta)^2]}$
- Cumulative distribution function $F_\theta(x) = \frac{1}{\pi} \arctan(x - \theta) + \frac{1}{2}$
- How to deal with continuous probability distribution?
- Quantize the distribution into equiprobable bins!
- Given $k \in \mathbb{N}$ create k bins $B_j = (F_\theta^{-1}(\frac{j-1}{k}), F_\theta^{-1}(\frac{j}{k})]$, $j = 1, \dots, k$



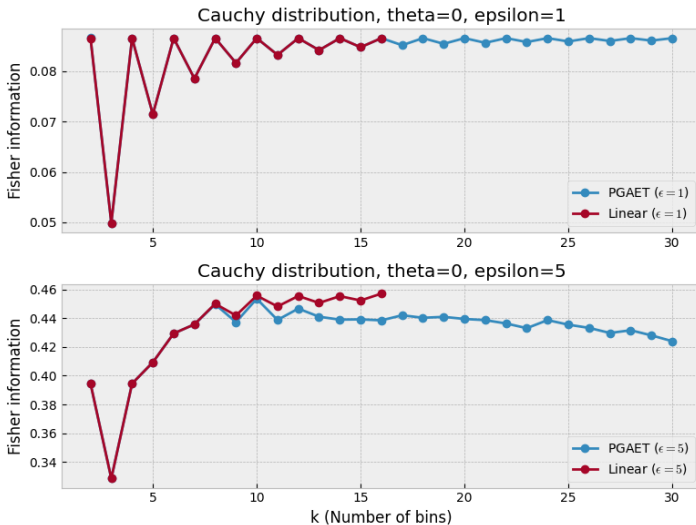
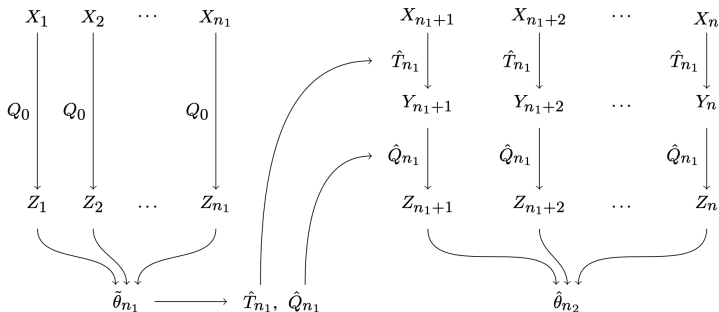


FIG 1. Graphical representation of the efficient two-step α -sequentially interactive privacy mechanism and estimation procedure.



- Differential Privacy is an emerging framework of data protection
- Given ϵ -differentially private requirement, many mechanisms satisfy the constraints
- We want to find the mechanism that maximizes Fisher information $\max_{Q \in D_{\epsilon,k}} I_{\theta}(Q)$
- Previous work constrained by the exponential time complexity $O(2^k)$
- Application of gradient ascent scales polynomially $O(k^9)$
- Challenges remain in low privacy regimes